

Martin's Life Work

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Chapter 1

Calculus

We start with Real Analysis. This loosely following [1]

1.1 Set Theory

The elements of a set can be chosen abstractly. For instance, in a university, a set of people can contain students, teachers, and faculty. We denote the objects of the set as generalized things. For instance, numbers, animals, and personal emotions are all things that we can place in a set.

Denote the set by the curly-brackets $\{\}$. The following operations all

Definition 1.1.

- Let $\cup$ define set union. Ie, if $x \in A, y \in B, \{x, y\} \in A \cup B$.
- Let $\cap$ define set intersection. This is defined as if $x, y \in A \cap B$, then $x \in A, x \in B, y \in A, y \in B$.
- Let A be a set. Let B be a subset of A , that is, $B \subset A$. Then, let B^c denote the compliment of B , or the points in A that are not in B . If $A = B$, then $B^c = \emptyset$.

We can use these operations to discuss some interesting groups of numbers. The following are a few useful sets.

Definition 1.2. The following are commonly used sets;

- Natural numbers: $\mathbb{N} = \{1, 2, 3, \dots\}$
- Whole Numbers: $\mathbb{W} = \{0, 1, 2, 3, \dots\}$
- Integers: $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
- Rational Numbers: $\mathbb{Q} = \left\{ \frac{x}{y} : \{x, y\} \in \mathbb{Z}, y \neq 0 \right\}$
- Irrational Numbers: $\mathbb{P} = \{x : x \text{ cannot be written as a fraction.}\}$
- Real Numbers: $\mathbb{R} = \mathbb{Q} \cup \mathbb{P} = \mathbb{Q} \cup \mathbb{Q}^c$

1.1.1 Properties of Set Operations

The operations of set union and intersection allow us to learn about the properties of sets under union, intersection, and compliment. We can define a few interesting theorems or results that allow us to start getting some logic in the world.

Theorem 1.1.

$$X^c \cap Y^c = (X \cup Y)^c \tag{1.1}$$

Proof. To prove this, we need to show that the LHS is equal to the RHS. Consider $X \cup Y$. If $x \in X^c$ then $x \notin X$. If $y \in Y^c$, $y \notin Y$. The overlap of the compliment sets, $X^c \cap Y^c$, contains neither x nor y . Thus, $x, y \in X \cup Y$, This completes the proof as since $x, y \notin X^c \cap Y^c$, then we must have that $(X \cup Y)^c = X^c \cap Y^c$. ■

For a union or intersection of many sets, define the notation;

$$\bigcup_{i=1}^{\infty} X_i = X_1 \cup X_2 \cup \dots, \quad \bigcap_{i=1}^{\infty} X_i = X_1 \cap X_2 \cap \dots$$

Theorem 1.2.

$$\left(\bigcap_{i=1}^{\infty} X_i \right)^c = \bigcup_{i=1}^{\infty} X_i^c \quad (1.2)$$

Proof. Let $x_i \in X_i$, $x_i \notin X_i^c$. Split this into two cases, where $\bigcap X_i \neq \emptyset$ and $\bigcap X_i = \emptyset$.

- $\bigcap X_i = \emptyset$, then for every x_i there exists a disjoint set X_j which doesn't contain x_i . This implies that $x_i \in X_j^c$.
- $\bigcap X_i \neq \emptyset$. If the intersection is not empty, then there exists $x_i \in X_1, X_2, \dots$. Thus, since x_i is simultaneously contained within every intersected set; $x_i \notin \bigcup X_i^c$.

This completes the proof stating that the compliment of the intersection of all sets will contain all points outside of the intersection which is the union of each individual compliment. ■

1.1.2 Functions

We can furthermore define functions as sets by addressing rules. Depending on these rules, a function can map an element in set A to set B . This is defined using the following notation;

$$f : A \rightarrow B \quad (1.3)$$

Example 1.1. Suppose $f(x) = 3x^2$ for $x \in \mathbb{R}$. Define the set $\mathbb{R}_+ = [0, \infty)$. Thus, we would say that;

$$f : \mathbb{R} \rightarrow \mathbb{R}_+ \quad (1.4)$$

We can classify functions based on the uniqueness of their mappings. For a function $f : A \rightarrow B$, we assume that each element in A maps to at most one element in B . A general function takes an element in A and maps to at most one value in B . Not all elements need to be mapped to, and some points can be visited multiple times. **Injectivity** implies that each input on A produces a unique result in B . Ie, there are no duplicates. However, not all elements in B are necessarily covered. **Surjectivity** implies that all outputs in B are mapped to, however, multiple $x \in A$ may map to elements in B . A map is **bijective** if every point in A uniquely maps to a point in B . This is also known as a set which is injective and surjective. See Fig. 1.1 for a visualization.

There are some useful notation. Here, for $f : X \rightarrow Y$, define the **image** as the set $Y = f(X)$. similarly, define the inverse image, or **preimage** as the set $X = f^{-1}(Y)$.

Definition 1.3. general functions have the following properties when used in tandem with set union and intersection. In addition, we present the inverse operations.

- $f(A \cup B) = f(A) \cup f(B)$
- $f(A \cap B) \subseteq f(A) \cap f(B)$
- $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$
- $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$
- $f^{-1}(f(A)) \supseteq A$
- $f(f^{-1}(C)) \subseteq C$

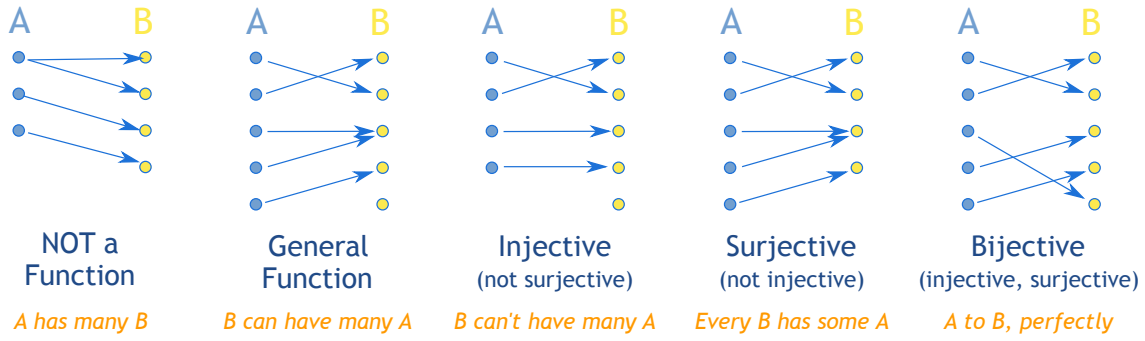


Figure 1.1: Function types based on correspondance. For a function $f : A \rightarrow B$, we assume that each element in A maps to at most one element in B . A general function takes an element in A and maps to at most one value in B . Not all elements need to be mapped to, and some points can be visited multiple times. **Injectivity** implies that each input on A produces a unique result in B . Ie, there are no duplicates. However, not all elements in B are nessecarily covered. **Surjectivity** implies that all outputs in B are mapped to, however, multiple $x \in A$ may map to elements in B . A map is **bijective** if every point in A uniquely maps to a point in B .

Proof. We aim to prove each statement mentioned above. Let $f : X \rightarrow Y$ and $A, B \subset X$, $C \subset Y$.

- a. Let $x \in f(A \cup B)$. Thus, there exists a y such that $y \in A \cup B \rightarrow x \in f(y)$ Since $y \in A, \rightarrow x \in f(A)$ or $y \in B \rightarrow x \in f(B)$. Thus, $f(A \cup B) \subseteq f(A) \cup f(B)$. Similarly, in the converse, suppose that $x \in f(A)$. then, there will exists a $y \in A$. Since $y \in A, y \in (A \cup B)$. Thus, $f(A) \subseteq f(A \cup B)$. Or more relevantly, $f(A) \cup f(B) \subseteq f(A \cup B)$. The converse with B is trivial. Thus, $f(A \cup B) = f(A) \cup f(B)$.
- b. Let $x \in f(A \cap B)$. There exists a y such that $y \in A$ and $y \in B$, so $y \in A \cap B$. So, if $y \in A$, $x \in f(A)$ and $y \in B$, $x \in f(B)$. If both are true simultaneously, then $f(A \cap B) \subseteq f(A) \cap f(B)$.
- c. Assume that $x \in f^{-1}(A \cup B)$ Then, $f(x) \in A \cup B$. Since $f(x) \in A$ and $f(x) \in B$ and for all A, B there exists a preimage, then $f^{-1}(A \cup B) \subseteq f^{-1}(A) \cup f^{-1}(B)$. The second inclusion is shown by the converse. Let $x \in f^{-1}(A) \cup f^{-1}(B)$. Since $f(x) \in A \cup B$ then $f^{-1}(A) \cup f^{-1}(B) \subseteq f^{-1}(A \cup B)$. Both inclusions are true, so there is an equality.
- d. Let $y \in f^{-1}(A \cap B)$ such that $f(y) \in A \cap B$. So, $f(y) \in A$ and $f(y) \in B$. Thus, $y \in f^{-1}(A)$ and $y \in f^{-1}(B)$. Or $y \in f^{-1}(A) \cap f^{-1}(B)$. So, $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$.
- e. Let $x \in f^{-1}(f(A))$. The inverse image of $f(A)$ will point to A if f is injective. Otherwise, it is at most a subset of A . $f^{-1}(f(A)) \supseteq A$
- f. Let $x \in f(f^{-1}(A))$. The image of the preimage is the identity if every element in A is mapped to. Thus, the identity is equal if f is surjective.

■

Remark that if f is a bijective map, then $f^{-1}(f(A)) = f(f^{-1}(A))$.

1.1.3 Problems

Problem 1.1. Let $\leq, \geq$ have their regular definitions. Let $A = \{0 \leq x \leq 3, x \in \mathbb{R}\}, B = \{-1 \leq x \leq 1, x \in \mathbb{R}\}$. Show that $A \cap B = C = \{0 \leq x \leq 1, x \in \mathbb{R}\}$

Solution. To show this, we need to show that the LHS is equal to the RHS for the final line in the problem statement. Trivially, $C \subset A$; $\inf C \in A, \sup C \in A$. Thus, $\forall x \in C, x \in A$. Similarly, $C \subset B$ by the same logic. Thus, state $C^c = (-\infty, 0) \cup (1, \infty)$. Furthermore, writing the union of elements in the compliments of A, B ; $A^c \cup B^c = (-\infty, 0) \cup (1, \infty)$. We use the property of intersections of sets

$$\begin{aligned} A^c \cup B^c &= (A \cap B)^c \\ C^c &= (A \cap B)^c \\ C &= A \cap B \end{aligned}$$

■

Problem 1.2. List elements of;

$$(\{2, 3, 4\} \cup \{x \in \mathbb{R} : x^2 - 4x + 3 = 0\}) \cap \{x \in \mathbb{R} : -1 \leq x < 3\} \quad (1.5)$$

Solution. The LHS of this contains two sets. The right-set are solutions to a quadratic equation. This contains two elements.

$$x_{1,2} = \frac{4 \pm \sqrt{16 - 12}}{2} = \{1, 3\}$$

Thus, the LHS is the union of two sets. Rewriting this;

$$\{1, 2, 3, 4\} \cap \{x \in \mathbb{R} : -1 \leq x < 3\} \quad (1.6)$$

The intersection of the discrete set of integers between 1 and 4 and the open neighborhood between -1 and 3 is simply $\{1, 2\}$ as 3 is not included in the set.

$$(\{2, 3, 4\} \cup \{x \in \mathbb{R} : x^2 - 4x + 3 = 0\}) \cap \{x \in \mathbb{R} : -1 \leq x < 3\} = \{1, 2\} \quad (1.7)$$

Problem 1.3. If $A \subset S$, show the following;

- $(A^c)^c = A$
- $A \cup A = A \cap A = A \cup \emptyset = A$
- $A \cap \emptyset = \emptyset$
- $A \times \emptyset = \emptyset$

Solution.

- Let $x \in A^c$. $x \notin A$. Thus, $(A^c)^c = A$.
- If $x \in A$ and $x \in A$ and A is the same set, logically, the union of both sets is just the set. Similarly, the intersection with itself is just itself as all points are covered within the same set. The set $\emptyset$ is defined as the set which contains nothing. Thus, $A \cup \emptyset$ is the union of the set A with nothing. Since $\emptyset$ doesn't add anything new to the set, one logically writes that $A \cup \emptyset = A$.
- Let $x \in A$. Since $x \notin \emptyset$ then $x \notin A \cap \emptyset$. However, nothing intersects with the emptyset, so naturally, $A \cap \emptyset = \emptyset$ as there is nothing to intersect over.
- The cartesian product of the emptyset is again defined by the property of the emptyset having no elements. Since there is no ordered pair formed when taking the cartesian product with the emptyset, the logical output set is a set with no elements, which is the emptyset.

Problem 1.4. Prove the following statements:

1. $A^c \cup B^c = (A \cap B)^c$
2. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
3. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Problem 1.5. We prove each statement from top to bottom. To show equalities, we need to show that the LHS equates the the RHS.

Let $x \in A \cap B$. Since x is in both A and B , then $x \notin A^c \cup B^c$. Thus, since x is general; $A^c \cup B^c = (A \cap B)^c$. Similarly, let $x \in A^c \cup B^c$, then by logical, x can be every value within except the points which overlap in A and B as their complements do not touch points which are contained in both sets.

Let $x \in A \cap (B \cup C)$ This set is in both A and $(B \cup C)$. Thus, $x \in A, B \cup C \rightarrow, A, B \rightarrow, x \in (A \cap B)$. $x \in A, B \cup C \rightarrow, A, C \rightarrow, x \in (A \cap C)$. So, x must be in one or the other. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Let $x \in A \cup (B \cap C)$, or x is in A or B and C . Thus, we have two inclusions, $x \in A \cup B$, or its in A or B and $A \cup C$, or A or C . Since x must at least be in A , the left term is easy to see, however, if $x \in (B \cap C)$ it must be in both $B \cap C$. Thus, $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Problem 1.6. Show the following;

$$B \cap \left(\bigcup_{i=1}^{\infty} X_i \right) = \bigcup_{i=1}^{\infty} B \cap X_i$$

$$\left(\bigcup_{i=1}^{\infty} X_i \right)^c = \bigcap_{i=1}^{\infty} X_i^c$$

Solution.

1. Let $x \in B \cap (\cup X_i)$. Since $x \in \cup X_i$ there exists at least one j such that $x \in X_j$. If $x \in B$ and $x \in X_j$ then there exists at least one non-zero entry such that $x \in B \cap X_j$. Thus, since x is general, we have that $B \cap (\cup X_i) = \cup (B \cap X_i)$.
2. Let $x \in \cup X_i$. There exists that for at least one integer j , $x \in X_j$. Thus, $x \notin X_j^c \rightarrow x \notin \cap X_j^c$. Since x is general, we have that $(\cup X_i)^c = \cap X_i^c$.

1.2 The Real Number System

The real numbers are defined by a set of axioms. There are various ways to treat this. I am going to just state all of the properties upfront and then proceed with the consequences of their existence.

1.2.1 Axioms and Order

Theorem 1.3. Axioms of the Real Number System: *These are the axioms that define the set of real numbers. The properties are stated outright, then the consequences are listed in further statements.*

F1: Commutativity; Let $a, b \in \mathbb{R}$, then $a + b = b + a$, and $a \cdot b = b \cdot a$.

F2: Associativity; Let $a, b, c \in \mathbb{R}$, then $(a + b) + c = a + (b + c)$, and $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.

F3: Distributivity; $a \cdot (b + c) = ab + ac$

F4: Existence of Neutral Numbers; $a + 0 = a$, $a \cdot 1 = a$

F5: Existence of Additive and Multiplicative Inverse Numbers; For any $a \in \mathbb{R}$ there exists a number b such that $a + b = 0$, where $b = -a$. Similarly, there exists a number c such that $a \cdot c = 1$, where $c = a^{-1}$.

O6: Order; There is a subset of $\mathbb{R}$ known as $\mathbb{R}_+$ such that the two properties are satisfied:

- If $a, b \in \mathbb{R}_+$, $a + b \in \mathbb{R}_+$, $a \cdot b \in \mathbb{R}_+$.
- For any $a \in \mathbb{R}$ only one of the following is true.

$$a \in \mathbb{R}_+, \quad a = 0, \quad -a \in \mathbb{R}_+$$

The numbers such that $a \in \mathbb{R}_+$ are known as positive numbers, while the numbers such that $-a \in \mathbb{R}_+$ are known as negative numbers.

LUB7: Least Upper Bound; A nonempty set of real numbers that is bounded from above will have a least upper bound.

The real number system is completely defined by the Thm. 1.3. We can now form the properties and rules of algebra on the real number system.

1. In a sum or product of many numbers, the parentheses can be omitted; $(a + (b + c)) + d = a + ((b + c) + d) = (a + b) + (c + d) = \dots$ by F1 and F2.
2. In a product of many numbers, order is immaterial; $a \cdot b \cdot c = b \cdot a \cdot c = c \cdot a \cdot b = \dots$
3. For any $a, b \in \mathbb{R}$ the equation $x + a = b$ has one and only one solution. For, if $x \in \mathbb{R}$ then see the following computation.

$$\begin{aligned} x &= x + 0 \\ &= x + ((a) + (-a)) \\ &= (x + a) + (-a) \\ &= b + (-a) \\ x &= b + (-a) \end{aligned}$$

Thus, $x = b + (-a)$ is the only solution.

4. For any $a, b \in \mathbb{R}$, with exception $a \neq 0$ the equation $xa = b$ has one and only one solution.

$$\begin{aligned} x &= xaa^{-1} \\ &= (xa) \cdot a^{-1} \\ x &= ba^{-1} \end{aligned}$$

Thus, the element 1 is unique from F4. we define $a^{-1} = \frac{1}{a}$

5. For any element $a \in R$, $a \cdot 0 + a \cdot 0 = a \cdot (0 + 0) = a \cdot 0 = a \cdot 0 + 0$. So both $a \cdot 0$ and 0 are solutions of the equation $x + a \cdot 0 = 0$. Thus, if a product of numbers is 0 then at least one of the element must be 0 . for if $ab = 0$ and $a \neq 0$ then we can multiply both sides by a^{-1} to get $b = 0$. Hence, the illegitimacy of division by 0 .
6. Solutions of the equation $x + (-a) = 0$ are given by $-(-a)$ given by (3.). By F5, we have $-(-a) + (-a) = 0$, so $-(-a) = a$ for any $a \in \mathbb{R}$.
7. $(a^{-1})^{-1} = a$ for any non-zero $a \in \mathbb{R}$. Since $a \cdot a^{-1} = 1$ by F5, and we know that $a^{-1} \neq 0$, then F4 implies that $(a^{-1})^{-1}$ and a are equal as both are solutions to $x + a^{-1} = 1$.
8. $-(a + b) = (-a) + (-b)$ for all $a, b \in \mathbb{R}$ as both are solutions to the equation $x + (a + b) = 0$.
9. $(ab)^{-1} = a^{-1}b^{-1}$ for any $a, b \in \mathbb{R}$, $a, b \neq 0$. Both are solutions to the equation $x(ab) = 1$.

From this, we can define the rules of the fraction;

$$\begin{aligned} \frac{ac}{bc} &= ac(bc)^{-1} = ab^{-1}cc^{-1} = ab^{-1} = \frac{a}{b} \\ \frac{a}{b} \cdot \frac{c}{d} &= ab^{-1}cd^{-1} = ac(bd)^{-1} = \frac{ac}{bd} \\ \frac{a}{b} + \frac{c}{d} &= ab^{-1} + cd^{-1} = ab^{-1}dd^{-1} + cbb^{-1}d^{-1} \\ &= (ad + cb) \cdot (b^{-1}d^{-1}) = \frac{ad + cb}{bd} \end{aligned}$$

10. $a \cdot (-1) = -a$ for all $a \in \mathbb{R}$. Since $(-1) \cdot a + a = a((-1) + 1) = a \cdot 0 = 0$. We have that $(-1) \cdot a$ and a are solutions to the equation $x + a = 0$, hence are equal.

A consequence of this is that $-a \cdot b = (-1) \cdot a \cdot b = -ab$ and $(-a) \cdot (-b) = (-1) \cdot (-1) \cdot a \cdot b = ab$

These are the field properties of the real numbers and apply to fields such as the rational and complex numbers. To define the reals, The order and LUB properties need to be applied. We can now discuss the order properties of the real numbers.

1. If $a, b \in \mathbb{R}$ only one of the following statements is true;

$$\begin{aligned} a &< b \\ a &= b \\ a &> b \end{aligned}$$

This is the case as we apply $(a - b)$ then the number must fit in one of three subsets, being $a \in \mathbb{R}_+$, $-a \in \mathbb{R}_+$, $a \in \{0\}$.

2. if $a > b$ and $b > c$ then $a - b \in \mathbb{R}_+$ and $b - c \in \mathbb{R}_+$. Thus, $a - c = (a - b) + (b - c) \in \mathbb{R}_+$. So, $a > c$.
3. If $a > b$ and $c \geq d$, then $a + c \geq b + d$. This implies that $a - b \in \mathbb{R}_+$, $c - d \in \mathbb{R}_+ \cup \{0\}$. Thus, $(a + c) - (b + d) = (a - b) + (c - d) \in \mathbb{R}_+$.
4. If $a > b > 0$ and $c \geq d > 0$, then $ac > bd$. For $a - b \in \mathbb{R}_+$ and $c \in \mathbb{R}_+$, then $ac - bc = (a - b)c \in \mathbb{R}_+$. Similarly, $c - d \in \mathbb{R}_+ \cup \{0\}$, $b \in \mathbb{R}_+$, and so $bc - bd \in \mathbb{R}_+ - \{0\}$. Thus, it follows that $ac - bd = (ac - bc) + (bc - bd) \in \mathbb{R}_+$, that is, $ac > bd$.

5. Rules for addition and multiplication of real numbers;

- $(a_+) + (a_+) = (a_+)$
- $(a_-) + (a_-) = (a_-)$
- $(a_+) \cdot (a_+) = (a_+)$
- $(a_-) \cdot (a_-) = (a_+)$

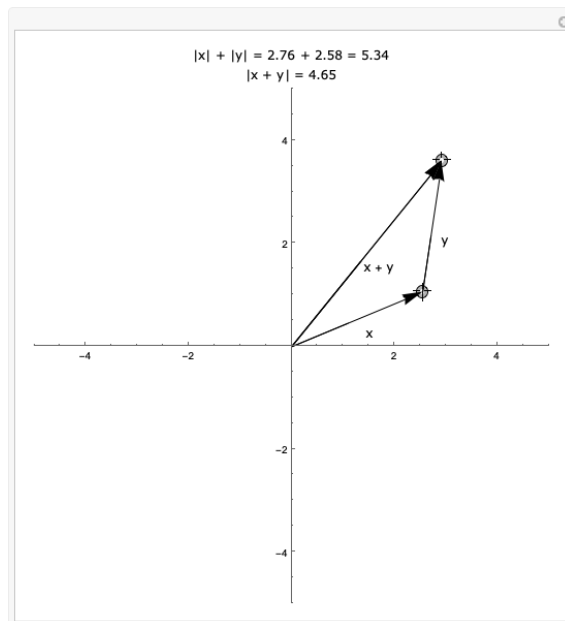


Figure 1.2: visualization of the triangle inequality; $|x + y| \leq |x| + |y|$ for all $x, y \in \mathbb{R}$.

- $(a_+) \cdot (a_-) = (a_-)$
- 6. For any $a \in \mathbb{R}, a^2 \in \mathbb{R}_+$. More tightly, $a^2 \geq 0$ where the equality is $0^2 = 0$ and special exceptions are $1^2 = 1$.
- 7. If $a > 0$, then $1/a > 0$.
- 8. If $a > b > 0$ then $1/b > 1/a$.

We can introduce exponentiation from the consequent multiplication of identical elements.

$$a^n = a \cdot a \cdot a \cdot \dots$$

1.2.2 Absolute Value

The absolute value is defined as the following piecewise function;

$$|a| = \begin{cases} a & \text{if } a > 0 \\ 0 & \text{if } a = 0 \\ -a & \text{if } a < 0 \end{cases}$$

The absolute value has several properties;

1. $|a| \geq 0$ for all $a \in \mathbb{R}$ and $|a| = 0$ iff. $a = 0$.
2. $|ab| = |a| \cdot |b|$ for all $a \in \mathbb{R}$.
3. $|a|^2 = a^2$ for all $a \in \mathbb{R}$.
4. $|a + b| \leq |a| + |b|$ for all $a, b \in \mathbb{R}$.
5. $|a - b| \geq ||a| - |b||$ for all $a, b \in \mathbb{R}$.

The triangle inequality (4.) is of utmost importance. See in Fig. 1.2.

Proof. The proof of (4.) will be finished. ■

1.2.3 Least Upper Bound Property.

The final concept for the real number system is the property of the least upper bound (LUB). If $S \subset \mathbb{R}$, then an upper bound for the set S is a number $a \in \mathbb{R}$ such that $s \leq a$ for each $s \in S$. If such a number exists, S is bounded from above. We call a real number y a **least upper bound** of the set S if y is an upper bound for S , and if a is any given upper bound for S , then $y \leq a$.

A non-empty, finite subset $S \subset \mathbb{R}$ will always have a LUB. We need to justify *LUB7*, which states that a non-empty set of real numbers, bounded from above, will always have a LUB.

The consequences of the LUB are listed;

1. For any real number x , there is an integer such that $n > x$.

Proof. By contradiction, suppose there is a number that doesn't obey this. Then, $n \leq x$ and thus is bounded from above. Since the set of integers is non-empty and bounded from above by the number a . However, for each integer n , $n + 1$ is also an integer. So, $n + 1 \leq a$. Thus, $n \leq a - 1$ is also an upper bound. Since $a - 1 < a$, $a - 1$ is not an upper bound, hence this is a contradiction. ■

2. For any positive number a there exists an integer n such that $\frac{1}{n} < a$.

Proof. Since $a \in \mathbb{R}$, by the field property, $a^{-1} = 1/a \in \mathbb{R}$. Since we prove that for all real numbers, there exists $n > x$ we have that there exists integer $j > a$ such that $1/a > 1/j$ by the order property. Hence, $1/j > 0$. ■

3. For any $x \in \mathbb{R}$, there is an integer n such that $n \leq x < n + 1$.

Proof. Let N be an integer such that $|N| > x$. Then, we have that x is bounded by the inequality $-N < x < N$. We have that this forms the set $\{-N, -N + 1, \dots, 0, 1, 2, \dots, N\}$. We can choose n arbitrarily along this such that $n \leq x < n + 1$. ■

4. For any $x \in \mathbb{R}$ and positive integer N there is an integer n such that

$$\frac{n}{N} < x < \frac{n+1}{N}$$

Proof. We can first rewrite this equation by multiplying each set by N ,

$$n < Nx < n + 1$$

We need to show that $N \cdot x \in \mathbb{R}$. Since $x \in \mathbb{R}$ and $N \in \mathbb{Z} \subset \mathbb{R}$ we have that $x \cdot N \in \mathbb{R}$ by the field properties. Thus, let $y = Nx \in \mathbb{R}$, by (3.) we have

$$n \leq y < n + 1$$

■

5. if $x, a \in \mathbb{R}, a > 0$ then there exists a rational number r such that $|x - r| < a$. That is, we can approximate a real number as closely as we wish by a rational number.

Proof. We first use (2.) to find a positive integer n such that $\frac{1}{n} < a$. By (4.) we can use that for x there exists $\frac{n}{N} < x < \frac{n+1}{N}$. Rewriting this by subtracting n/N ;

$$0 < x - \frac{n}{N} < \frac{1}{N} < a$$

Thus, We have that $|x - \frac{n}{N}| < a$ ■

1.2.4 Decimals

For a given set of integers, $a_1, a_2, \dots, a_n \in \{0, 1, 2, \dots, 9\}$, we can write the decimal representation as

$$a_0.a_1a_2a_3\cdots = \frac{a_0}{10^0} + \frac{a_1}{10^1} + \frac{a_2}{10^2} + \cdots + \frac{a_n}{10^n}$$

Suppose $n, m \in \mathbb{Z}, m < n$. Then, we have the inequality that;

$$\begin{aligned} a_0.a_1a_2\cdots a_m &\leq a_0.a_1a_2\cdots a_n \\ &= a_0.a_1a_2\cdots a_m + a_{m+1} \cdot 10^{-(m+1)} + \cdots + a_n \cdot 10^{-n} \\ &\leq a_0.a_1a_2\cdots a_m + 9 \cdot 10^{-(m+1)} + \cdots + 9 \cdot 10^{-n} \end{aligned}$$

If we add a 1 to the last entry, it rounds up to the decimal in question;

$$a_0.a_1a_2\cdots a_m \leq a_0.a_1a_2\cdots a_n < a_0.a_1a_2\cdots a_m + 10^{-m}$$

Formally, the infinite decimal $a_0.a_1a_2\cdots$ is a representation of the least upper bound. Ie, every infinite decimal is a decimal expansion for a definite real number and we can use the infinite decimal itself as a symbol for the number.

$$a_0.a_1a_2\cdots = \text{LUB}\{a_0.a_1a_2\cdots a_n : n \in \mathbb{Z}_+\} \quad (1.8)$$

and, for any positive integer n we have the inequality

$$a_0.a_1a_2\cdots a_n \leq a_0.a_1a_2\cdots \leq a_0.a_1a_2\cdots a_n + 10^{-n}$$

Using this, we can write numbers that are not rational.

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1.2.5 Square Roots

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1.2.6 Problems

Chapter 2

Measure Theory

2.1 Measurable Spaces

2

2.1.1 Problems

Problem 2.1. *Partition generated by a σ -algebras.*

1. Let $\mathcal{C} = \{A, B, C\}$ be a partition of E . List all the elements of $\sigma\mathcal{C}$
2. let $\mathcal{C}$ be a countable partition of E . show that every element of $\sigma\mathcal{C}$ is a countable union of elements taken from $\mathcal{C}$.
3. Let $E = \mathbb{R}$. Let $\mathcal{C}$ be the collection of all singleton subsets of $\mathbb{R}$, that is, sets that only contain a single point. Show that each element of $\sigma\mathcal{C}$ is either countable, or the complement of a countable set.

Solution.

1. The elements of $\sigma\mathcal{C}$ are defined as the intersection of all sigma-algebras that contain $\mathcal{C}$. Since $\mathcal{C}$ contains 3 subsets of E , as it is a partition, which implies that A, B, C are disjoint and have a union of E . Thus, we can define $\sigma\mathcal{C}$ as the following set:

$$\sigma = \{\emptyset, E, \{A\}, \{B\}, \{C\}, \{A \cap B\}, \{A \cap C\}, \{B \cap C\}\} \quad (2.1)$$

This set contains E , all partitions of closures by E generated from $\mathcal{C}$. Thus, this is the smallest sigma-algebra that contains E generated by $\mathcal{C}$.

2. Since $\mathcal{C}$ is a partition, $\bigcup \mathcal{C} = E$. We need to show that $\sigma\mathcal{C}$ contains only elements from $\mathcal{C}$. We can take ε as the set of all countable unions taken from E . We can show that this is a sigma-algebra by taking its definitions. Since ε contains E the subset of all the partitions, it must also contain the union of none of the partitions, or $\emptyset$. Secondly, by taking all of the unions, we start with the disjoint partitions and then create their unions until we reach E . This will be closed under complement as by taking countable unions, we will have the sets that are not part of the union as another subset of a different union. Thus, we can argue that this is the same as $\sigma\mathcal{C}$ by the logic of construction. Thus, since each element of $\sigma\mathcal{C}$ is isomorphic to the construction that we obtained from the union of subsets, we have shown that all elements are just a set of unions in $\mathcal{C}$.

3. Define the set $\mathcal{A} = \{A \subset \mathbb{R} : A \text{ is countable or } A^c \text{ is countable}\}$. We want to show that this is a sigma-algebra. Since $\emptyset$ is countable, it is contained within $\mathcal{A}$. $\emptyset \in \mathcal{A}$. The complement of this is the set of reals; $\mathbb{R} = \emptyset^c \in \mathcal{A}$. By definition, if A is countable, then A^c is co-countable. Similarly, if A is co-countable, this implies that A^c is countable. Thus, both sets are contained in $\mathcal{A}$.

Thus, consider a sequence $(A_n)_{n \in \mathbb{N}} \subset \mathcal{A}$. the union of this sequence is $\bigcup_n A_n$. This is split into cases. First, if all A_n are countable, so it the union, which is included in A_n . If all A_n are co-countable, then consider the compliment of the union $(\bigcup_n A_n)^c = \bigcap_n A_n^c$ which is an intersection of countable sets, which is countable. If there is a mix between countable and co-countable sets, they are co-countable, which are contained in A by the previous definition. Thus, $\bigcup_n A_n \in \mathcal{A}$. Thus, $\mathcal{A}$ is closed under union and complement. Thus, $\mathcal{A}$ is a sigma-algebra. Thus, $\sigma\mathcal{C} \subset \mathcal{A}$

We want to show that any countable set is a union of the singletons. Since A contains countable and co-countable sets, A is a union of singletons, which are by definition contained in $\mathcal{A}$.

$$A = \bigcup_{x \in A} \{x\} \quad (2.2)$$

For co-countable sets in A , we note that the complements are countable, which are contained in the above definition. Since $\mathcal{A}$ consists of countable sets, these are the countable unions of singletons contained in $\sigma\mathcal{C}$. Since $\mathcal{A}$ also consists of co-countable sets, these are the complements of intersections of countable sets in $\sigma\mathcal{C}$. Thus, $\mathcal{A} \subset \sigma\mathcal{C}$.

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Problem 2.2. Let (E, ε) be a measurable space. Fix $D \subset E$ and let

$$\mathcal{D} = \varepsilon \cap D = A \cap D : A \in \varepsilon$$

Show that $\mathcal{D}$ is a sigma-algebra on D . It is called the trace of (E, ε) on D .

Solution. To show that $\mathcal{D}$ is a sigma-algebra, we need to show closure under union, complement, intersection, and the existence of $D, \emptyset \in \mathcal{D}$.

1. To show that $D, \emptyset \in \mathcal{D}$ i will write out the logical steps. Since $D \subset E$, $D \in \varepsilon$. Since $A = D$, $D \cap D = D$. By definition of $\mathcal{D}$, $D \in \mathcal{D}$. Similarly, $\emptyset \subset E$, $\emptyset \in \varepsilon$. Since $A = \emptyset$, $\emptyset \cap D = \emptyset$. By definition of $\mathcal{D}$, $\emptyset \in \mathcal{D}$.
2. The closure of complement can be shown by the set property of D^c . Since $E^c = \emptyset$, $D^c = E \cap D$.

$$\begin{aligned} (A_n \cap D)^c &= A^c \cup D^c \\ &= A^c \cup (E \cap D) \\ &= E \cap (D^c \cup A) \\ &\in \mathcal{D} \end{aligned}$$

Thus $(A_n \cap D)^c \in \mathcal{D}$.

3. For union and intersection we will use a similar set operation;

$$\begin{aligned} \left(\bigcup_{i=1}^{\infty} A_i \right) \cap D &= \lim (A_1 \cup A_2 \cup \dots \cup A_n) \cap D \\ &= \lim (A_1 \cap D) \cup (A_2 \cap D) \cup \dots \cup (A_n \cap D) \\ &= \bigcup_{i=1}^{\infty} (A_i \cap D) \\ &\in \mathcal{D} \end{aligned}$$

$$\begin{aligned}
\left(\bigcap_{i=1}^{\infty} A_i\right) \cap D &= (A_1 \cap A_2 \cap \cdots \cap A_n) \cap D \\
&= \lim (A_1 \cap D) \cap (A_2 \cap D) \cap \cdots \cap (A_n \cap D) \\
&= \bigcap_{i=1}^n (A_i \cap D) \\
&\rightarrow \left(\bigcap_{i=1}^{\infty} (A_i \cap D)\right)^c \\
&= \bigcup_{i=1}^{\infty} (A_i \cap D)^c \\
&= \bigcup_{i=1}^{\infty} (A_i^c \cup D^c)
\end{aligned}$$

By the closure of the complement: $\in \mathcal{D}$

Thus we have shown that $\mathcal{D}$ is a sigma-algebra on D .

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Problem 2.3. Let (E, ε) be a measurable space, let $\{\Delta\}$ be a single-point not in E . Let $\bar{E} = E \cup \{\Delta\}$. Show that

$$\hat{\varepsilon} = \varepsilon \cup \{A \cup \{\Delta\} : A \in \varepsilon\}$$

.....

Problem 2.4. Let $(E, \varepsilon), (F, \mathcal{F})$ be measurable spaces. Show that the product $\varepsilon \otimes \mathcal{F}$ is also the sigma-algebra generated by $\hat{\varepsilon} \cup \mathcal{F}$ where

$$\hat{\varepsilon} = \{A \times F : A \in \varepsilon\} \quad \mathcal{F} = \{E \times A : A \in \mathcal{F}\} \quad (2.3)$$

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2.2 Measureable Functions

Measureable functions are defined as maps between Measureable spaces. Let $(E, \mathcal{E})$ and $(F, \mathcal{F})$ be measurable spaces. A measurable function is defined as

$$f : E \rightarrow F, \quad f^{-1}B \in \mathcal{E} \quad \forall B \in \mathcal{F}$$

2.2.1 Problems

Problem 2.5. Let E be a set and $(F, \mathcal{F})$ a measurable space. For $f : E \rightarrow F$, define

$$f^{-1}\mathcal{F} = \{f^{-1}B : B \in \mathcal{F}\}$$

Where $f^{-1}B = \{x \in E : f(x) \in B\}$. Show that $f^{-1}\mathcal{F}$ is a sigma-algebra on E .

Solution. Since $f^{-1}\mathcal{F} = \{f^{-1}B : B \in \mathcal{F}\}$ we need to show the properties. Since $f^{-1}F = E$, $E \in f^{-1}\mathcal{F}$. Similarly $f^{-1}\emptyset = \emptyset$, $\emptyset \in f^{-1}\mathcal{F}$. To show complement, consider $A \in \mathcal{F}$. Then $A^c \in \mathcal{F}$. Thus, we require by definition that $f^{-1}A^c \in E$. Thus the set is closed under complement for all $A \in \mathcal{F}$. For Union and intersection we use the definition;

$$\left(\bigcup_{i=1}^{\infty} A_i\right) \in \mathcal{F}$$

$$f^{-1}\left(\bigcup_{i=1}^{\infty} A_i\right) \in E$$

$$\left(\bigcap_{i=1}^{\infty} A_i\right)^c = \bigcup_{i=1}^{\infty} A_i^c$$

$$\in \mathcal{F}$$

$$f^{-1}\bigcap_{i=1}^{\infty} A_i \in E$$

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2.3 Measures

Let $(E, \mathcal{E})$ be a measurable space. A measure on $(E, \mathcal{E})$ is defined as a map $\mu : \mathcal{E} \rightarrow \bar{\mathbb{R}}_+$ where $\bar{\mathbb{R}}_+ = [0, +\infty]$. The measure has properties;

$$\mu(\emptyset) = 0$$

$$\mu\left(\bigcup_n A_n\right) = \sum_n \mu(A_n) \text{ for every disjointed sequence } (A_n) \in \mathcal{E}$$

The definition of the measure contains several properties;

1. If $A \cap B = \emptyset$ Then $\mu(A \cup B) = \mu(A) + \mu(B)$
2. If $A \subset B$ Then $\mu(A) \leq \mu(B)$
3. If $A_n \nearrow A$ Then $\mu(A_n) \nearrow \mu(A)$
4. $\mu(\bigcup_n A_n) \leq \sum_n \mu(A_n)$

Some examples of measures are listed below for the measurable space $(E, \mathcal{E})$;

1. Dirac Measure:

$$\delta_x(A) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

2. Counting Measure: Let D be a fixed subset of E . The counting measure counts the number of points $A \cap D$. We interpret this as all elements outside D have zero measure and all elements of D have equal measure.

$$\#_D(A) = \sum_{x \in D} \delta_x(A), \quad A \in \mathcal{E}$$

3. Discrete Measure: Let D be a countable subset of E , and $m(x)$ a positive number.

$$\mu(A) = \sum_{x \in D} m(x) \delta_x(A), \quad A \in \mathcal{E}$$

2.3.1 Problems

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2.4 Integration

Let $(E, \mathcal{E}, \mu)$ be a measure space. Here we define $\mathcal{E}$ as the sigma algebra of all $\mathcal{E}$ -measurable functions, which are all measurable functions mapping to the Borel sigma-algebra.

We define the 'integral of f with respect to μ ' as:

$$\mu f = \mu(f) = \int_E \mu(dx) f(x) = \int_E f d\mu \quad (2.4)$$

Definition 2.1. Let f be simple and positive, such that it has the form $f(x) = \sum_{i=1}^n a_i 1_{A_i}(x)$. Then we define

$$\mu f = \sum_{i=1}^n a_i \mu(A_i)$$

This definition can be used on different measures to see how the integral is defined.

1. Dirac Measure:

$$\mu f = \delta_x(A) f(x) = f(A)$$

2. Discrete Measure:

$$\mu f = \sum_{x \in D} m(x) f(x)$$

3. Lebesgue Integrals:

$$\mu f = \text{Leb}_E f = \int_E \text{Leb}_E(dx) f(x) = \int_E dx f(x)$$

Note that if the lebesgue exists, so does the riemann integral, which is reminded by the notation in the last equality. Note that the converse is false.

The main idea of the integral with respect to a measure is packaged in how the measure acts on infinitesimal pieces dx . Since the measure gives different weights to different parts of the set. We can use a result that approximates a measurable function f as a convergent sequence of simple functions. That is, we can approximate f using simple functions that partition the domain in a way that converges to the value of the integral.

Chapter 3

Probability

3.1 Combinatorics

Problem 3.1. *Given N people in a room, what is the probability that 2 or more people share a birthday?*

Solution. This problem is an exercise in counting. First, let's go case-by-case and write out the general solution. There are 365 days in total. If there is only 1 person in the room, only 1 day is taken. Once we place the second person in the room, we can calculate the probability of the two individuals not-sharing the same birthday;

$$\mathbb{P}[\text{share birthday}] = 1 - \mathbb{P}[\text{not share birthday}] \quad (3.1)$$

I.e., the second person has only 364 days to choose from to not share. Thus, their probability of not sharing a birthday are $364/365$, i.e. the event space over the sample space. Likewise, the following person has now only 363 days to choose from and so on. This relation can be multiplied together such that the overall probability of not sharing is that nobody in the room is sharing. This can be built using a product;

$$\mathbb{P}[\text{not share birthday}] = \prod_{x=1}^N \frac{365 - x}{365} \quad (3.2)$$

This gives us the probability that at least 2 people are sharing;

$$\mathbb{P}[\text{at least 2 share birthday}] = 1 - \prod_{x=1}^N \frac{365 - x}{365} \quad (3.3)$$

Plotting this relation as a function of N , we can see the shape of this relation in Fig. 3.1.

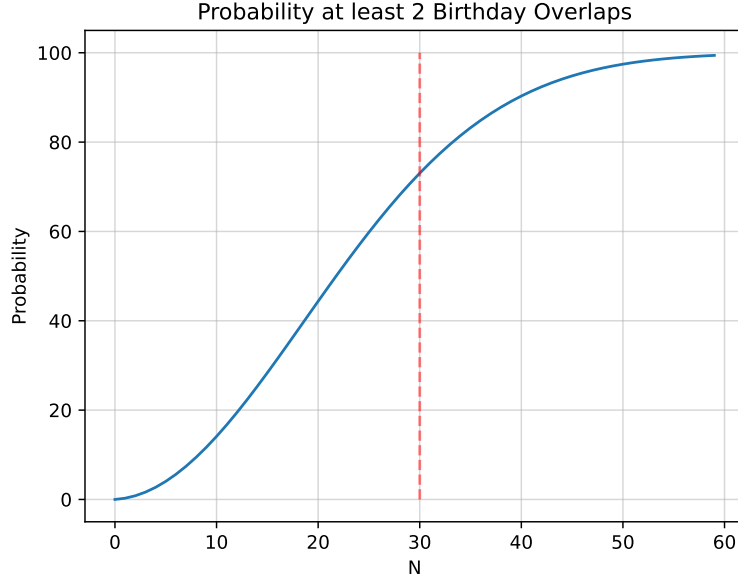


Figure 3.1: Probability of at least 2 people in a room sharing a birthday as a function of N people in the room. at $N = 30$, $\mathbb{P} \approx 73\%$

3.2 Multivariate Distributions

Let $\mathbf{X}$ be a random vector with components $X_1, \dots, X_n$. The distribution is arrives from is $P(x_1, x_2, \dots, x_n)$.

The marginal distribution is defined as

$$P(x_1, \dots, x_s) = \int P(x_1, \dots, x_s, x_{s+1}, \dots, x_r) dx_{s+1} \dots dx_r$$

where the left hand side is defined as the conditional distribution. It is often written in terms of Baye's rule, as well as the set defined probability;

$$P(x_1, \dots, x_s | x_{s+1} \dots x_r) = \frac{P(x_1, \dots, x_s, x_{s+1}, \dots, x_r)}{P(x_{s+1}, \dots, x_r)} \quad (3.4)$$

And written in set notation;

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (3.5)$$

The moments of the multivariate distribution are defined as

$$\left\langle \prod_{i=1}^n X_i^{m_i} \right\rangle = \int x_1^{m_1} x_2^{m_2} \dots x_n^{m_n} P(x_1, x_2, \dots, x_n) dx_1 dx_2, \dots dx_n$$

The characteristic function occurs as the product of Fourier transforms;

$$\begin{aligned} G(k_1, \dots, k_n) &= \left\langle e^{i(k_1 X_1 + \dots + k_n X_n)} \right\rangle \\ &= \sum_{i=0}^{\infty} \frac{(ik_1)^{m_1} (ik_2)^{m_2} \dots (ik_n)^{m_n}}{m_1! m_2! \dots m_n!} \left\langle \prod_{i=1}^n X_i^{m_i} \right\rangle \end{aligned}$$

Where one extracts the coefficients from the expansion. The log of this function extracts the cumulants, known as the cumulant generating function;

$$\log[G(k_1, \dots, k_n)] = \sum_{i=0}^{\infty} \frac{(ik_1)^{m_1} (ik_2)^{m_2} \dots (ik_n)^{m_n}}{m_1! m_2! \dots m_n!} \left\langle \prod_{i=1}^n X_i^{m_i} \right\rangle_c \quad (3.6)$$

3.3 Covariance Matrix

The covariance matrix is defined as the $n \times n$ matrix $\langle X_i X_j \rangle_c$ Which stands for the second cumulant, or the variance.

$$\langle X_i X_j \rangle_c = \langle X_i X_j \rangle - \langle X_i \rangle \langle X_j \rangle \quad (3.7)$$

When normalized, this matrix forms the correlation coefficients between the random variables:

$$\rho_{i,j} = \frac{\langle X_i X_j \rangle - \langle X_i \rangle \langle X_j \rangle}{\sqrt{(\langle X_i^2 \rangle - \langle X_i \rangle^2)(\langle X_j^2 \rangle - \langle X_j \rangle^2)}} \quad (3.8)$$

3.3.1 Transformations

Transformations of random-variables answer the question of given X with PMF/PDF f_X , how does one calculate the distribution of some $Y = r(X)$? The ben-br notes define a simple process;

Theorem 3.1. *The transformation of a random variable can be given by the following procedure;*

- For each y , find the set $A_y = \{x : r(x) \leq y\}$
- Find the CDF of y

$$F_Y(y) = \int_{A_y} f_X(x) dx \quad (3.9)$$

- The PDF is the derivative of the CDF;

$$f_Y(y) = F'(y) \quad (3.10)$$

For transformations that are strictly monotone, then there is a well-defined inverse, $s = r^{-1}$, then the process is a simple formula;

$$f_Y(y) = f_X(s(y)) \left| \frac{ds(y)}{dy} \right| \quad (3.11)$$

Example 3.1. *Let $X \sim \text{Exp}(\lambda)$ and $r = \xi^2$. Then how can we find the transformation $Y = r(X)$?*

1. Using the general method, we can calculate the PDF exactly;

$$\begin{aligned} p(Y \leq y) &= p(r(X) \leq y) \\ &= p(X^2 \leq y) \\ &= p(X \leq \sqrt{y}) \\ F_Y(y) &= 1 - e^{-\lambda\sqrt{y}} \\ f_Y(y) &= \frac{\lambda}{2\sqrt{y}} e^{-\lambda\sqrt{y}} \end{aligned}$$

2. Since $r(X)$ is a monotonic function with an inverse $s(x) = \sqrt{x}$ then we can compute the formula directly;

$$\begin{aligned} f_Y(y) &= \lambda e^{-\lambda\sqrt{y}} \cdot \left| \frac{1}{2\sqrt{y}} \right| \\ &= \frac{\lambda}{2\sqrt{y}} e^{-\lambda\sqrt{y}} \end{aligned}$$

Thus, both methods yield the correct transformed PDF.

3.3.2 Expectations

The expectation of a random variable is given by the explicit integral over the CDF. Specifically, we are interested in the value that is most likely to occur. This can be formalized by adding all of the cumulative probability over the range of values. Here we assume that X is non-negative, which eases the bounds of integration. Otherwise, this can be recast to have a negative sign to ensure all regions are positive;

$$\int_0^\infty p(X \geq x)dx = \int_0^\infty \left(\int_x^\infty f_Y(y)dy \right) dx$$

This integral sweeps a region in xy space. This region is fixing x from $0 \rightarrow \infty$. Then, y integrates from x to infinity;

$$0 \leq x, \quad x \leq y \leq \infty$$

Thus, we can integrate instead over x which is from 0 to y , then overall all space;

$$\begin{aligned} \int_0^\infty \left(\int_x^\infty f_Y(y)dy \right) dx &= \int_0^\infty \left(\int_0^y f_Y(y)dx \right) dy \\ &= \int_0^\infty y f_Y(y)dy \\ &:= \mathbb{E}[y] \end{aligned}$$

Thus, the expectation is defined more commonly by the law of the unconscious statistician;

Theorem 3.2. *Expectation of random variable or function of a random variable is defined by a CDF of a random variable;*

$$\mathbb{E}[r(x)] = \int_{\Omega} r(x) f_X(x) dx \tag{3.12}$$

3.3.3 Characteristic Functions

Chapter 4

Stochastic Processes

4.1 Point Processes

Define a point process by the following rules. The sample space is consisting of a set of states which consist of $s = 0, 1, 2, \dots$. For each s , there is a set of s real numbers τ which is increasing.

The probability distribution of these states is given by a sequence of non-negative functions Q_s . The normalization is given by;

$$1 = Q_0 + \int_{-\infty}^{\infty} d\tau_1 Q_1(\tau_1) + \int_{-\infty}^{\infty} d\tau_1 \int_{-\infty}^{\infty} d\tau_2 Q_2(\tau_1, \tau_2) + \dots \quad (4.1)$$

It is common to permit only symmetric functions of $Q_s(t_1, \dots, t_s)$ such that there are $s!$ permutations for each Q_s . Then the normalization condition becomes simple to write as a series;

$$1 = Q_0 + \sum_{s=1}^{\infty} \frac{1}{s!} \int_{-\infty}^{\infty} Q_s(t_1, \dots, t_s) dt_1 \cdots dt_s \quad (4.2)$$

Here it appears that $Q_s(t_1, \dots, t_s)$ is known as the probability of the sequence of events where s dots occur at some vector t_1 . Here, t can denote a time in a temporal process, as well as spatial coordinates.

For example, consider an ideal gas. The ideal gas has a volume in $\mathbb{R}^3$. Such a system can be defined by the unique property of each state, which in this case, can be the energy of the system. The number of discrete microstates is given by the relation to entropy

$$S = k_b \ln \Omega$$

Thus, we can find the number of microstates

$$\Omega = e^{S/k_b}$$

the probability of a single state, which consists of a state variable is proportional by definition of probability;

$$P(N, U, P, V) = Q_s(U) = \frac{1}{\Omega} = e^{-S/k_b} = e^{-\beta(U+PV-\mu N)} \quad (4.3)$$

This is defined as the Janossy density for a single state. The normalization condition is

$$\begin{aligned} 1 &= \sum_{n=0}^N e^{-\beta(U+PV-\mu n)} \\ &= e^{-\beta(U+PV)} \sum_{n=0}^N e^{\beta\mu n} \\ &= e^{-\beta(U+PV)} \frac{1 - e^{\beta\mu N}}{1 - e^{\beta\mu}} \end{aligned}$$

Another example is for a general counting process. Suppose that $Q_s(t_1, \dots, t_s)$ is the probability of exactly s detections at intervals dt_s . We can compute the average number of dots by defining

$$N(t) = \sum_{j=1}^s \mathbf{1}_{t_{j-1} < t < t_j}$$

. This just counts the number of dots by summing each interval by an indicator function.

$$\begin{aligned} \langle N(t) \rangle &= \sum_{s=0}^{\infty} \frac{1}{s!} \int_{-\infty}^{\infty} dt_1 \dots dt_s Q_s(t_1, \dots, t_s) \sum_{j=1}^s \mathbf{1}_{t_{j-1} < t < t_j} \\ s \text{ symmetric integrals over } [t_a, t_b] &= \sum_{s=1}^{\infty} \frac{s}{s!} \int_{t_a}^{t_b} dt_1 \int_{-\infty}^{\infty} \dots dt_s Q_s(t_1, \dots, t_s) \\ &= \sum_{s=1}^{\infty} \frac{1}{(s-1)!} \int_{t_a}^{t_b} dt_1 \int_{-\infty}^{\infty} \dots dt_s Q_s(t_1, \dots, t_s) \end{aligned}$$

And similarly for the second moment,

$$\begin{aligned} \langle N(t)^2 \rangle &= \sum_{s=0}^{\infty} \frac{1}{s!} \int_{-\infty}^{\infty} dt_1 \dots dt_s Q_s(t_1, \dots, t_s) \left[\sum_{i=1}^s \mathbf{1}_{t_{i-1} < t < t_i} \right] \left[\sum_{j=1}^s \mathbf{1}_{t_{j-1} < t < t_j} \right] \\ &= \sum_{s=0}^{\infty} \frac{1}{s!} \int_{-\infty}^{\infty} dt_1 \dots dt_s Q_s(t_1, \dots, t_s) \left[\sum_{i=j} \mathbf{1}_{t_{j-1} < t < t_j} + \sum_{i \neq j} \mathbf{1}_{t_{j-1} < t < t_j} \right] \\ s \text{ cases where } (i=j) \rightarrow \langle N \rangle &+ \sum_{s=0}^{\infty} \frac{1}{s!} \int_{-\infty}^{\infty} dt_1 \dots dt_s Q_s(t_1, \dots, t_s) \left[\sum_{i \neq j} \mathbf{1}_{t_{j-1} < t < t_j} \right] \\ s \text{ ways to fix } i, (s-1) \text{ fix } j &= \langle N \rangle + \sum_{s=2}^{\infty} \frac{s(s-1)}{s!} \int_{t_a}^{t_b} dt_1 \int_{t_a}^{t_b} dt_2 \int_{-\infty}^{\infty} dt_3 \dots dt_s Q_s(t_1, \dots, t_s) \\ &= \langle N \rangle + \sum_{s=2}^{\infty} \frac{1}{(s-2)!} \int_{t_a}^{t_b} dt_1 \int_{t_a}^{t_b} dt_2 \int_{-\infty}^{\infty} dt_3 \dots dt_s Q_s(t_1, \dots, t_s) \end{aligned}$$

Thus, we have recovered the variance of a general process being that of a Poisson process;

$$\langle N \rangle = \langle N^2 \rangle - \langle N \rangle^2$$

This is suprising as we haven't said anything about how Q_s is distributed!

4.1.1 Poisson processes

Random dot events are independent if the Janossy Density factorizes into the form of an independent joint probability density;

$$Q_s(t_1, \dots, t_s) = e^{-\nu} p(\tau_1) \dots p(\tau_s) \quad (4.4)$$

In this case, one can see that if we define

$$\langle N \rangle = \nu = \int_{t_a}^{t_b} p(\tau) d\tau$$

we find the following result.

$$\begin{aligned} \langle e^{ikN} \rangle &= e^{-\nu} \sum_{s=1}^{\infty} \frac{1}{s!} \left[\int_{\infty}^{\infty} e^{ik \sum_{\sigma} 1_{t_{\sigma}} p(\tau) d\tau} \right]^s \\ &= \exp \left[\int_{\infty}^{\infty} e^{ik \sum_{\sigma} 1_{t_{\sigma}} p(\tau) d\tau} - \nu \right] \\ &= \exp \left[\int_{\infty}^{\infty} (e^{ik \sum_{\sigma} 1_{t_{\sigma}}} - 1) p(\tau) d\tau \right] \\ \text{surviving terms are only in } [t_a, t_b] &= \exp \left[(e^{ik} - 1) \int_{t_a}^{t_b} p(\tau) d\tau \right] \end{aligned}$$

This characteristic function is precisely that of the Poisson distribution, notably that the probability of N dots falling in an interval $[t_a, t_b]$ is;

$$P(N) = \frac{\langle N \rangle^N}{N!} e^{-\langle N \rangle} \quad (4.5)$$

This result is that an independent waiting time, or an independent process probability gives rise to a Poisson distributed process.

4.2 Stochastic Variables and Properties

As known, we can define a stochastic process for a given random variable and some function. Typically, if this is a value of time, it is known as a general stochastic process.

$$Y_X(t) = f(X, t) \quad (4.6)$$

For example, assume the form $Y(t) = \psi(t)X$. Assume that here $\psi(t) > 0$ such that the math works out nicely, otherwise, the probability inequality will switch signs.

$$Y(t) = \psi(t)X \quad (4.7)$$

If we know the distribution of X , one can write that

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(x) dx \quad (4.8)$$

Thus, we can calculate the CDF of the transformed random variable;

$$\begin{aligned} P(Y \leq y) &= P(\psi(t)X \leq y) \\ &= F_X \left(\frac{1}{\psi(t)} y \right), \quad \psi(t) > 0 \end{aligned}$$

The requirement on $\psi(t)$ is a valid one, as if the function is ever equal to 0, then the process is not random as it is simply 0 for that time. We can then calculate the PDF of Y ;

$$\begin{aligned} f_Y(y) &= \frac{d}{dy} F_X \left(\frac{1}{\psi(t)} y \right) \\ &= \frac{1}{|\psi(t)|} f_X \left(\frac{1}{\psi(t)} y \right) \end{aligned}$$

Expectations

We can take expectations as normal on the random variable. It is interesting to see how the expectations collapse;

$$\langle Y(t) \rangle = \int_{-\infty}^{\infty} Y_x(t) P_X(x) dx \quad (4.9)$$

For the above example, we find a the same solution using the definition of the underlying distribution and from the transformed one;

$$\begin{aligned} \langle Y(t) \rangle &= \psi(t) \int_{-\infty}^{\infty} x f_X(x) dx \\ &= \int_{-\infty}^{\infty} y f_Y(y) dy \\ &= \int_{-\infty}^{\infty} \psi(t) x \frac{1}{|\psi(t)|} f_X \left(\frac{1}{\psi(t)} y \right) dy \\ &= \int_{-\infty}^{\infty} x f_X(x) \psi(t) dx \\ &= \psi(t) \int_{-\infty}^{\infty} x f_X(x) dx \end{aligned}$$

Autocorrelation

As a reminder, the auto-correlation is defined as

$$\kappa(i, j) = \langle (X_i - \langle X_i \rangle)(X_j - \langle X_j \rangle) \rangle \quad (4.10)$$

so, for a stochastic process, the function is a function of two time points;

$$\begin{aligned} \kappa(t_1, t_2) &= \langle (Y(t_1) - \langle Y(t_1) \rangle)(Y(t_2) - \langle Y(t_2) \rangle) \rangle \\ &= \langle Y(t_1)Y(t_2) \rangle - \langle Y(t_2) \rangle \langle Y(t_1) \rangle \end{aligned}$$

Once again, for the separable example above, we see that the time dependent functions factor out;

$$\kappa(t_1, t_2) = \psi(t_1)\psi(t_2) \left[\langle x^2 \rangle - \langle x \rangle^2 \right]$$

4.3 Stationary Processes

As we write earlier, a general stochastic process Y depends on some variables and some random variables. A stationary stochastic processes is one that has a stationary, or non-unique variance and mean along the temporal variable. Typically, these are reserved to contain mean and variances that coincide with the time-step between time.

For example, consider two processes; the Wiener process, and the Ornstein-Uhlenbeck process.

The Wiener process is defined as the following,

$$W_t - W_s \sim \mathcal{N}(0, t - s)$$

probabilistically, this is describing the probability of moving to some W_t given some W_s . We can fully defined this process by defining the transition probability, as well as the initial distributive probability;

$$P_1(y_1, t) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{y_1^2}{2t}}$$

$$P_{1|1}(y_2, t_2 | y_1, t_1) = \frac{1}{\sqrt{2\pi(t-s)}} e^{-\frac{(y_2 - y_1)^2}{2(t-s)}}$$

This is compared to the Ornstein-Uhlenbeck process, which is notably defined as Markov and Stationary, with an autocorrelation of $e^{-\tau}$. Let $\tau = t - s$;

$$P_1(y_1, t) = \frac{1}{\sqrt{2\pi}} e^{-\frac{y_1^2}{2}}$$

$$P_{1|1}(y_2, t_2 | y_1, t_1) = \frac{1}{\sqrt{2\pi(1 - e^{-2\tau})}} e^{-\frac{(y_2 - y_1)^2}{2(1 - e^{-2\tau})}}$$

As one can see, the OH process is stationary as it forgets its initial position, whilst the Wiener process has a memory of this, hence, it is not markovian.

4.4 Power Spectra

4.5 Probabilistic Hierarchy

The probabilistic Hierarchy is defined as a way to form a joint distribution for a stochastic process. Specifically, we can write down the following statement;

$$P(y_1, t_1; y_2, t_2; \dots; y_n, t_n) = \int \delta(y_1 - Y_x(t_1)) \delta(y_2 - Y_x(t_2)) \dots \delta(y_n - Y_x(t_n)) p_X(x) dx \quad (4.11)$$

Here, $Y_x(t)$ is a realization of a process. We are then integrating over this realization weighted by the probability of each random value.

Example 4.1. Suppose that $Y_x(t) = X \cos(\omega t)$ for $X \sim \mathcal{N}(0, 1)$. Then, we can find $P(y_1, t_1)$

$$P(y_1, t_1) = \int \delta(y_1 - Y_x(t_1)) P_X(x) dx$$

$$= \int \delta(y_1 - x \cos(\omega t)) \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

Using the fact that

$$\delta(y_1 - x \cos(\omega t)) = \frac{1}{\cos(\omega t)} \delta\left(x - \frac{y_1}{\cos(\omega t)}\right)$$

or

$$= \int \delta(y_1 - x \cos(\omega t)) \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

$$= \frac{1}{\cos(\omega t)} \int \delta\left(x - \frac{y_1}{\cos(\omega t)}\right) \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

$$= \frac{1}{\cos(\omega t)} \frac{1}{\sqrt{2\pi}} e^{-\left(\frac{y_1}{\cos(\omega t)}\right)^2/2}$$

Thus,

$$P(y_1, t_1) = \frac{1}{\cos(\omega t)} \frac{1}{\sqrt{2\pi}} e^{-\frac{(y_1)^2}{2 \cos^2(\omega t)}} \quad (4.12)$$

4.6 Markov Processes

Markov processes are defined by a set of probabilistic equations.

4.7 Chapman-Kolmogorov Equations

4.8 Problems

Problem 4.1. Find the mean-squared distance after N steps on a random walk on a $2D$ square lattice when U -turns are forbidden.

Solution. The transition probability of this problem requires we know of the current and previous locations. Let n be the (x, y) position at some time, and m be the position at the previous time. Let $\xi_1 = n_1 - m_1$ be the difference of the two times. Since the steps have equal probability, we can write the transition matrix of this problem. Let $\uparrow, \downarrow, \rightarrow, \leftarrow$ denote the vectors of length 1 moving in the respective directions.

$$\begin{aligned} T(\xi_2|\xi_1) = & \frac{\delta(\xi_1 - \rightarrow)}{3} [\delta(\xi_2 - \leftarrow) + \delta(\xi_2 - \uparrow) + \delta(\xi_2 - \downarrow)] \\ & + \frac{\delta(\xi_1 - \leftarrow)}{3} [\delta(\xi_2 - \rightarrow) + \delta(\xi_2 - \uparrow) + \delta(\xi_2 - \downarrow)] \\ & + \frac{\delta(\xi_1 - \uparrow)}{3} [\delta(\xi_2 - \rightarrow) + \delta(\xi_2 - \leftarrow) + \delta(\xi_2 - \downarrow)] \\ & + \frac{\delta(\xi_1 - \downarrow)}{3} [\delta(\xi_2 - \rightarrow) + \delta(\xi_2 - \uparrow) + \delta(\xi_2 - \leftarrow)] \end{aligned} \quad (4.13)$$

Given this transition, we can calculate the either the characteristic function or moments explicitly. Since there aren't that many delta functions to calculate, we can directly calculate the moments. From the definition of ξ , we have that;

$$\vec{x}_n = \vec{x}_0 + \sum_{j=1}^N \vec{\xi}_j \quad (4.14)$$

Thus, assume that $x_0 = 0$ and we can compute moments around $\vec{x}$ instead of ξ .

$$\begin{aligned} \langle \vec{x}^2 \rangle &= \left\langle \left[\sum_{j=1}^N \vec{\xi}_j \right]^2 \right\rangle \\ &= \left\langle \left[\sum_{j=1}^N \vec{\xi}_j \right] \left[\sum_{i=1}^N \vec{\xi}_i \right] \right\rangle \\ &= \left\langle \sum_{i=1}^N \sum_{j=1}^N \vec{\xi}_i \vec{\xi}_j \right\rangle \\ &= \sum_{i=1}^N \langle \vec{\xi}_i^2 \rangle + 2 \sum_{i=1}^N \sum_{j=1}^{i-1} \langle \vec{\xi}_i \vec{\xi}_j \rangle \end{aligned} \quad (4.15)$$

We compute the moments explicitly. Note that $\rightarrow + \leftarrow = 0$; $\uparrow + \downarrow = 0$;

$$\begin{aligned} \langle \xi \rangle &= \sum_{\xi_1} \left[\sum_{\xi_2} \xi_2 T(\xi_2|\xi_1) \right] P(\xi_1) \\ &= \frac{1}{4} \left(\frac{1}{3}(\uparrow + \rightarrow + \leftarrow) + \frac{1}{3}(\downarrow + \rightarrow + \leftarrow) + \frac{1}{3}(\uparrow + \downarrow + \leftarrow) + \frac{1}{3}(\uparrow + \rightarrow + \downarrow) \right) \\ &= 0 \end{aligned}$$

This being said, we can compute the square moment and the correlation in time. It can be easily shown that since $\rightarrow \cdot \rightarrow^T = 1$, then $\langle \xi_i^2 \rangle = 1$. The correlation can be computed for jointwise distance $|i - j|$. We will use the recursive property to find the correlation.

$$\begin{aligned}\langle \xi_{j+1} | \xi_j \rangle &= \frac{1}{3} (\xi_j + \perp + (-\perp)) \\ &= \frac{1}{3} \xi_j\end{aligned}$$

Thus, from the recursive property of the correlation;

$$\langle \xi_{j+k} \xi_j \rangle = 3^{-k} \quad (4.16)$$

Then, the MSD is fully calculated;

$$\begin{aligned}\langle x^2 \rangle &= \sum_{i=1}^N \langle \xi_i^2 \rangle + 2 \sum_{i=1}^N \sum_{j=1}^{i-1} \langle \xi_i \xi_j \rangle \\ &= N + 2 \sum_{i=1}^N \sum_{j=1}^{i-1} \lambda^{i-j} \\ &= N + 2\lambda \sum_{k=0}^{N-1} (N-k) \lambda^k \\ &= N + 2\lambda \left[\frac{(N-1)(1-\lambda^N)}{1-\lambda} - \frac{\lambda - (N-1)\lambda^N + (N-1)\lambda^{N+1}}{(1-\lambda)^2} \right] \\ &= N + 2\lambda \left[\frac{N}{1-\lambda} - \frac{(1-\lambda^N)}{(1-\lambda)^2} \right] \\ &= \frac{N(1+\lambda)}{1-\lambda} - \frac{2\lambda(1-\lambda^N)}{(1-\lambda)^2}\end{aligned}$$

Thus, we have that the MSD after N steps is the equation above, for $\lambda = 1/3$;

$$\langle x^2 \rangle = \frac{N(1+\lambda)}{1-\lambda} - \frac{2\lambda^2(1-\lambda^N)}{(1-\lambda)^2} \quad (4.17)$$

Note that since $N(1+\lambda)/(1-\lambda) = 2N$ There is a higher order of diffusivity compared to the free case, which N .

Chapter 5

Electromagnetism

5.1 Debye-Huckel Theory

Debye-Huckel theory is a description of the electromagnetic interactions between ions in a solution of ionic compounds.

5.1.1 Poisson-Boltzmann

The potential in a field of ions must be solved in terms of electrostatics in a media. This is described by Poisson's equation

$$\nabla^2 \Psi(\mathbf{r}) = -4\pi \frac{\rho(\mathbf{r})}{\varepsilon} \quad (5.1)$$

In a sea of ions, we must carefully define the charge density as it pertains to the system. In a general solution of ions, there will be j species of ions.

$$\nabla^2 \Psi(\mathbf{r}) = -4\pi \frac{\rho_j(\mathbf{r})}{\varepsilon}$$

The charge distribution of a single ion concentration is essentially the density of the ionic positions. It would be tedious and unfeasible to count every ion's position exactly as we are assuming too many things. Rather, one can use a statistical ensemble of ions via a canonical ensemble, which is a good approximation here.

$$\rho_i(\mathbf{r}) = C_i e_0 Z_i \exp[-\beta Z_i e_0 \Psi(r)] \quad (5.2)$$

Here the energy of the system is defined as the potential energy scaled by the charge defined as $U = qV$. The full equation is known as the Poisson-Boltzmann equation, where $\Psi(\mathbf{r})$ is the potential for all space.

$$\nabla^2 \Psi(\mathbf{r}) = -\frac{1}{\varepsilon_0 \varepsilon_r} \sum_{i=1}^{\infty} C_i e_0 Z_i \exp[-\beta Z_i e_0 \Psi(r)] \quad (5.3)$$

There is a nice approximation we can make given that the potential of the solution is small. We linearize the exponential to first order here.

$$\begin{aligned}
\nabla^2 \Psi(\mathbf{r}) &\approx -\frac{1}{\varepsilon_0 \varepsilon_r} \sum_{i=1}^{\infty} C_i e_0 Z_i [1 - \beta Z_i e_0 \Psi(r)] \\
&= -\frac{1}{\varepsilon_0 \varepsilon_r} \sum_{i=1}^{\infty} [C_i e_0 Z_i - \beta C_i Z_i^2 e_0^2 \Psi(r)] \\
&= -\sum_{i=1}^{\infty} \left[\frac{C_i e_0 Z_i}{\varepsilon_0 \varepsilon_r} - \kappa^2 \Psi(r) \right]
\end{aligned}$$

Where κ_i^2 is related to the debye-length, a unit of the electrostatic screening caused by the ionic interactions. The Debye length is defined as $\lambda_D = \kappa^{-1}$.

$$\kappa_i^2 = \frac{\beta C_i Z_i^2 e_0^2}{\varepsilon_0 \varepsilon_r} \quad (5.4)$$

5.1.2 Monovalent Ionic Mixture

Suppose there is a single type of ion in the mixture. The sum collapses into a single term and the constant term is dropped.

$$\nabla^2 \Psi(\mathbf{r}) = \kappa^2 \Psi(\mathbf{r}) \quad (5.5)$$

This equation has an obvious solution which are the real exponentials;

$$\Psi(\mathbf{r}) = A e^{-\kappa r} + B e^{\kappa r} \quad (5.6)$$

Chapter 6

Optics

Problem 6.1. *Solution Provided by Martin Jasinski, Neil Jadav.*

Solution. (a) The input beam is a single wavelength of light at a fixed power. $I_{in} = |\vec{E}_0| = 1$ W/cm². The input beam will hit a beam splitter and split in half towards two mirrors. Each mirror has a respective reflectivity, which reduces its intensity. The light then passes back towards the beam splitter, splitting one more time before reaching the detector. We can first write down the all the waves of interest. First, note that $I = |\vec{E}|^2$ and that $r = \sqrt{R}$

$$I_{in} = |E_0 \cdot \hat{x}| = 1 \quad [\text{W}/\text{cm}^2] \quad (6.1)$$

$$I_{out} = |E_1 + E_2|^2 \quad (6.2)$$

$$= \left| E_0 \frac{r}{2} e^{i2kd_1} + E_0 \frac{1}{2} e^{i2kd_1} \right|^2 \quad (6.3)$$

$$= \left| E_0 \frac{0.9}{2} e^{i2kd_1} + E_0 \frac{1}{2} e^{i2kd_1} \right|^2 \quad (6.4)$$

$$(6.5)$$

[2 Intensity Form]

Knowing the form of the output intensity, we can write out the norm of the out intensity.

$$\begin{aligned} I_{out} &= |E_0|^2 (0.45^2 + 0.5^2 + 0.45 \cos(2k(d_1 - d_2))) \\ &= |E_0|^2 (0.4525 + 0.45 \cos(2k\Delta d)) \end{aligned}$$

[4 cosine Term]

Thus, we can see a minimum and a maximum based on the form of the cosine term. The minimum of cosine is -1 and the maximum is 1. In addition, we know from I_{in} that the total power $E_0^2 = 1$, so we can plug in the limits and obtain the min and max.

$$\boxed{\begin{aligned} I_{\max} &= 0.9025 \quad [\text{W}/\text{cm}^2] \\ I_{\min} &= 0.0025 \quad [\text{W}/\text{cm}^2] \end{aligned}} \quad (6.6)$$

[1 Answer]

[1 Units]

.....

(b) In the case of a second frequency aligning with the maximums at $\Delta d = 3\mu\text{m}$, we can inspect the cosine term and discover if there is any thing similar.

$$\begin{aligned} \cos(2k\Delta d) &= \cos\left(\frac{4\pi}{\lambda}\Delta d\right) \\ &= \cos(2\pi n), \quad n \in \mathbb{Z} \end{aligned}$$

[3 argument integer periodic]

Thus, we can solve the relation for λ_1 ;

$$n = \frac{2}{\lambda}\Delta d = \frac{6 \cdot 10^{-6}}{600 \cdot 10^{-9}} = \frac{1}{10^{-1}} = 10 \quad (6.7)$$

Neighboring frequencies with the **same** maxima at $\Delta d = 3 \cdot 10^{-6}$ can be found by varying n by ± 1 ;

$$\lambda_{n=11} = \frac{6 \cdot 10^{-6}}{11} = 545.45 \cdot 10^{-9} \quad (6.8)$$

$$\lambda_{n=9} = \frac{6 \cdot 10^{-6}}{9} = 666.\bar{6} \cdot 10^{-9} \quad (6.9)$$

[3 calculate nearby lambdas]

Thus, the closes wavelength from the original, ie

$$\arg \min_{\lambda_2} [\lambda_1 - \lambda_2]$$

$$\boxed{\lambda_2 = 545.45 \quad [\text{nm}]} \quad (6.10)$$

[1 Correct Answer]

.....

(c) Suppose $\lambda_1 \rightarrow \lambda_2$, what changes? We can see that the only wavelength dependent term is the argument of the cosine. This term changes the amount of Δd required to see the maxima/minima. This does **not** change the amount of signal that the detector measures. Thus, all answers except $\mathcal{A}$ are incorrect.

$$\boxed{\text{Correct Answer: } \mathcal{A}} \quad (6.11)$$

[5 Correct Answer]

Chapter 7

Digital Image Theory and Application

I have been working on images at my lab for quite some time now. This is a summary and documentation of the things i have learned while on the job.

7.1 Images as a Hilbert Space

At its core, images are taken by a camera array, which contains *pixels*. Each pixel is one point of the image, or a binning of millions of photons hitting the image, creating a smooth description of some object being photographed. Definitions of the image can be done in 2 ways; theoretical and practical.

Definition 7.1. Let $f(x, y, z)$ be a field in $\mathbb{R}^3$. Let $\hat{P}$ be the projection operator such that $\hat{P} : \mathbb{R}^3 \rightarrow L_2(\mathbb{R}^2)$. Then the digital image is defined as $g(x, y) = \hat{P}f(x, y, z)$.

the results of this defintion imply that the dimension z is integrated out. For instance, images of physical objects are the integration of the density function of space, $\rho(x, y, z)$, or when using a light emission system, the optical field density $I(x, y, z)$.

This defintion of the image lives in the vector-space of $\mathbb{R}^2$, which allows for a set of known transformations. Affine transformations are defined by a transformation matrix $\mathbf{T}$ and a shift $\vec{v}$.

$$f_T(\vec{x}) = f(\mathbf{T}^{-1}(\vec{x}) + \vec{b}) \quad (7.1)$$

For group transformations, we can define the action $(g \circ f)(\vec{x}) = f(g^{-1}\vec{x})$. This definition implies linearity for sums of scalar transformations of images;

$$g(af_1 + bf_2) = a(g \circ f_1) + b(g \circ f_2) \quad (7.2)$$

Furthermore, we can define the group as being part of $SO(2)$. This group includes the rotation matrices, which gives the norm-preservation property;

$$|g \circ f| = |f|_{L_2} \quad (7.3)$$

Naturally, the functions we produce want to have some sense of similarity between them. Ie, how similar are two images? To do this, we can re-use the idea of a generalized angle. Two images can be the same in terms of angle, but have different magnitudes. This can be interpreted as a measure in distance. From QM, we know that a square-integratable set of functions with angle and distance are defined in the Hilbert space.

Definition 7.2. Let $\mathcal{H}$ denote the Hilbert space; Elements of $\mathcal{H}$ must satisfy the following criteria. Let $a, b \in \mathcal{H}$:

1. Elements have an inner product: $\langle a, b \rangle$
2. Norm Property: $\langle a, a \rangle > 0$
3. Zero Property: $\langle a, a \rangle = 0$ if $x = 0$
4. Inner product commutation: $\langle a, b \rangle = \langle b, a \rangle^*$
5. Linearity for constants: $\langle a, e_1 b_1 + e_2 b_2 \rangle = e_1^* \langle a, b_1 \rangle + e_2^* \langle a, b_2 \rangle$
6. Angle: $A(a, b) = \langle a, b \rangle$
7. Distance: $d(a, b) = \sqrt{\langle a - b, a - b \rangle}$

The inner product of two functions is evaluated as their integral over the domain, hence;

$$\langle f(x, y), g(x, y) \rangle = \iint f g^* dx dy \quad (7.4)$$

From QM, we know about the existence of unitary operators. Below is a definition that we can use to define rotation operations.

Definition 7.3. Let U be a unitary operator acting on f . $U_\theta : L_2(\mathbb{R}^2) \rightarrow L_2(\mathbb{R}^2)$ such that the norm is preserved $|U_\theta f| = |f|$.

7.1.1 Rotational Symmetry

Let U_θ be a unitary rotation operator in $SO(2)$.

We can define exact symmetry as an identical copy due to rotations; $U_\theta f = f$. We know that our images will contain some noise; $U_\theta(f + \eta) = f + U_\theta \eta$. These disturbances will not be symmetrical.

Definition 7.4. Let the rotational symmetry be defined as how similar the image is to its rotated counterparts normalized by the norm of the image.

$$S(f) = \sup_{\theta} \frac{\langle f, U_\theta f \rangle}{|f|^2} = 1 - \inf_{\theta} \frac{|f - U_\theta f|}{|f|} \quad (7.5)$$

We can also define the maximum symmetry being $S(f) = 1$, whereas minimum symmetry, or a total orthogonal set, $S(f) = 0$.

This function is bounded on $[0, 1]$. The supremum clause defines the largest possible deviations from a perfect symmetry.

This definition uses the auto-correlation of the image with its rotational counter-part, however, integrates over the radial-ness of the image. For parts of the image that are only radially symmetric for a set of angles, we can define the measure over a radial average;

For instance, say that the image is symmetric over a wide range of angles, but contains portions where the symmetry is broken by a mis-alignment or blip, these would be integrated over in the symmetry measure.

Definition 7.5. Let the radial average be defined as the integral over angles θ_0 .

$$f_0(r, \Delta\theta, \theta_0) = \frac{1}{2\pi} \int_{\theta_0}^{\theta_0 + \Delta\theta} f(r, \theta) d\theta \quad (7.6)$$

Then, define the symmetry metric as the ratio between the norm of the image, and the norm of the radial averaged-portion;

$$S(r, \theta_0, \Delta\theta) = \frac{|f_0|^2}{|f|^2} \quad (7.7)$$

Chapter 8

Computer Science and Algorithms

This is mainly about algorithms as i dont know shit about computers

8.1 Dynamic Programming

Dynamic programming is something i dont understnad

8.1.1 2D Chain-Link Algorithm

This is text

Algorithm 1 Dynamic Programming for Best 2D Chain of Points

Require: $\text{min_idx} = \{\text{min_idx}[t]\}_{t=0}^{F-1}$, list of candidate positions per frame1: $\text{min_vals} = \{\text{min_vals}[t]\}_{t=0}^{F-1}$, list of candidate values per frame2: $w = (w_x, w_y)$, weighting for distance computation (default (1, 1))3: max_step , optional maximum allowed step distance**Ensure:** $\text{idx_chain}, \text{x_chain}, \text{y_chain}$, the indices, positions, and values of the optimal path4: Initialize $F \leftarrow |\text{min_idx}|$ 5: Initialize $\text{cost}[i] \leftarrow 0$ for all i in frame 06: Initialize $\text{prev} \leftarrow []$ ▷ to store predecessor indices7: **for** $t = 1$ to $F - 1$ **do**8: $A_x \leftarrow \text{min_idx}[t - 1]$, $B_x \leftarrow \text{min_idx}[t]$ 9: $A_y \leftarrow \text{min_vals}[t - 1]$, $B_y \leftarrow \text{min_vals}[t]$

10: Compute pairwise differences:

$$\Delta_x[i, j] = |A_x[i] - B_x[j]|, \quad \Delta_y[i, j] = |A_y[i] - B_y[j]|$$

11: Compute weighted Euclidean distances:

$$D[i, j] = \sqrt{(w_x \Delta_x[i, j])^2 + (w_y \Delta_y[i, j])^2}$$

12: **if** max_step is not None **then**13: Set $D[i, j] \leftarrow \infty$ if $D[i, j] > \text{max_step}$ 14: **if** no finite $D[i, j]$ exists **then**15: **return** empty path16: **end if**17: **end if**

18: Compute cumulative path cost matrix:

$$M[i, j] = \text{cost}[i] + D[i, j]$$

19: For each node j in frame t , find predecessor $p[j]$ minimizing $M[:, j]$:

$$p[j] = \arg \min_i M[i, j], \quad \text{new_cost}[j] = M[p[j], j]$$

20: Update $\text{cost} \leftarrow \text{new_cost}$ 21: Append p to prev 22: **end for**23: Initialize $j \leftarrow \arg \min_i \text{cost}[i]$ ▷ index of minimal total cost in last frame24: Initialize $\text{idx_chain} \leftarrow [0]^F$, set $\text{idx_chain}[F - 1] \leftarrow j$ 25: **for** $t = F - 2$ down to 0 **do**26: $\text{idx_chain}[t] \leftarrow \text{prev}[t][\text{idx_chain}[t + 1]]$ 27: **end for**

28: Compute chains:

$$\text{x_chain}[t] = \text{min_idx}[t][\text{idx_chain}[t]], \quad \text{y_chain}[t] = \text{min_vals}[t][\text{idx_chain}[t]]$$

29: **return** $\text{idx_chain}, \text{x_chain}, \text{y_chain}$

Bibliography

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